



Society of Petroleum Engineers

SPE-229449-MS

Identifying Simple Criteria to Assess the Risk of Mud Loss During Drilling Operations Nearby Faults

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229449-MS](https://doi.org/10.2118/229449-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

In a West Africa offshore oil field, the drilling campaign faces challenges due to numerous normal faults in the overburden and the reservoirs. Crossing these faults with a mud weight too high have led to severe losses that increased the non-productive time and the risk of well control issues. Anticipating mud losses near faults remains challenging, as the conductivity and sealing capacity of the mud in such context remains hard to predict.

To optimize the drilling performance in the faulted sections, we built an extensive database by analyzing 232 mud loss events on and nearby faults crossed by 29 wells. We complement the database with the lithology, the stresses acting on the fault planes and the mud weight while drilling, or equivalent circulating density, that generated the losses. 1D mechanical models were done for each well to obtain these stresses, then we used the Mohr-Coulomb failure criteria (MC) with the mud pressure while drilling to examine fault instability.

We show that the mud losses are related to the rock contacts at the fault-well intersections and the pressure of drilling mud. Without making prior assumptions on the fault mechanical properties, such as cohesion and friction angle, the latter are directly obtained from the effective-stress state with mud pressure during losses that clearly align on a Mohr-Coulomb type failure line. The inverted mechanical properties of the fault are compared to the intact rock properties by applying the same friction angle, but by degrading the cohesion by a constant factor. Lithologies at fault contacts, like shale/sandstone or sandstone/sandstone, are more prone to significant mud losses compared to shale/shale contacts where the MC failure criteria is met. Losses occur in 68% of fault contacts involving sandstones, compared to 50% in shale-on-shale contacts. Moreover, we observed that high mud pressure used to drill development wells were more prone to significant mud losses.

From this analysis, we developed a simple, yet robust, method using the Mohr-Coulomb failure criterion to predict fault reactivation by mud pressure during drilling operations. It refines mud weight strategies and well design to address faults and proactively improves drilling performance for upcoming wells.

Introduction

Mud loss is one of the biggest challenges in drilling operations, particularly in naturally fractured formations and complex geological structures such as heavily faulted regions. The volume and rate of mud loss can be

significant, and in attempts to mitigate this issue, operators often reduce mud weight (MW) or employ Lost Circulation Materials (LCM). However, these actions can lead to severe incidents, including compromised safety and increased risks of well loss. Mud loss, regardless of its volume, directly contributes to Non-Productive Time (NPT) and subsequent economic losses. Therefore, predicting the risk of mud loss near faults before drilling operations is crucial to avoid severe consequences and ensure operational efficiency. Faults are inherently complex geological structures that make it hard to predict mud losses. Fault zones are three-dimensional objects comprised of deformed rock and characterized by complex internal structures (Sibson, 1986). Indeed, faults observed on seismic surveys are often large, of several kilometers, spanning several well paths (Brem et al., 2019). Mature fault structures are composed of three main parts: the fault core, a low-permeability zone often filled with clay-rich gouge and breccia; the damage zone, surrounding the core and containing highly fractured rocks that can act as conduits for fluid flow; and the intact zone, which is the undeformed rock outside the damage zone (Sibson, 1977; Caine et al., 1996). While fault planes are observed seismic cross sections, the complex structures of mature faults are invisible.

The permeability of faults and fractures significantly impacts the occurrence of mud loss through seepage. Borehole data indicate that active faults are the most significant hydraulic conduits because they exhibit higher permeability compared to faults that are not optimally oriented within the current stress field (Barton et al., 1995). Additionally, experimental data on fractured cores show a strong decrease in permeability when confinement increases (e.g., Gutierrez et al., 2000; Frash et al., 2017). These studies introduced the concept of hydraulic aperture closure indicating a threshold pressure between 20-90 bars below which the permeability dramatically increases. In field scale experiments (LSBB, Tournemire and Mont Terri), fault transmissivity increases of several orders of magnitude with effective normal stress ($\sigma_n - P_p$) between 1- 100 bars depending on lithology (Hyman et al., 2016; Guglielmi et al., 2021). A significant increase in transmissivity may be a consequence of fault reactivation and fracture opening. However, fault gouge present in mature fault zones complicates the relationship between permeability and porosity, making it hard to set a clear pressure threshold that can be used in drilling operations (Wibberley et al., 2008).

The Fracture Closing Pressure (FCP) is commonly set as the maximum mud weight in drilling, but relying on it can be overly optimistic and cause significant fluid losses, particularly in faulted areas. FCP is the minimum horizontal stress (S_{hmin}) derived from the 1D mechanical earth model (1DMEM) and calibrated with leak-off tests (LOT) and extended leak-off tests (XLOT). Our experience based from post mortem analysis, along with previous studies, indicates that mud losses are often linked to the proximity of fault zones rather than solely to fracture pressure (Brem et al., 2019; Abd Rahim et al., 2019). Thus, a more nuanced approach is needed to account for the complex interactions between mud weight and fault behavior.

On the other hand, assuming cohesionless faults may result in overly conservative mud weight strategies, with a significantly reduced loss gradient. This approach often relies on the Mohr-Coulomb failure criteria, using conservative properties such as zero cohesion and low friction angle of around 20 degrees, typical for caprock shales. often creating impractically narrow mud weight windows (Brem et al., 2019; Abd Rahim et al., 2019). Furthermore, this assumption may not hold in areas with complex geological conditions, leading to complications in drilling. Thus, refining our understanding of fault characteristics and adopting a more balanced assessment of mechanical properties is essential for developing safe and more efficient drilling strategies.

The main objective of this study is to present a comprehensive postmortem analysis of the mud losses observed in proximity of faults in a heavily faulted oilfield, focusing on the relation between fluid pressure and formation stresses. Furthermore, we identify an operational criterion for optimizing drilling performance in analogous neighboring fields.

Geological and structural context

This study focuses on data from an offshore oil field in West Africa, where the drilling campaign encounters significant challenges due to the presence of numerous faults. The geology of the area is complex and has experienced various tectonic phases. Currently, the sedimentary basin remains active and contains a dense network of faults oriented in different azimuths.

The sedimentary basin has experienced deposition from the Oligocene to the Pleistocene, with the most significant deposits occurring during the Miocene epoch. It is primarily composed of deep marine siliciclastic sediments, which are now located at an average water depth of 1,500 meters. The reservoirs within this basin are characterized by a complex arrangement of turbiditic channels and lobes, all set within a matrix of marine shales. Notably, both the reservoir and overburden rocks are unconsolidated, which may impact their mechanical properties and behavior during drilling activities.

This region exhibits various faulting modes that reflect its major tectonic phases, with most faults being normal. Initially, large thrust faults formed during a compressional phase, marked by signs of compression in the basement. An extensional phase followed, typical of a passive margin, during which deltaic sediments were deposited in the marine environment. This stage introduced a complex stress field that led to the formation of regional transtensional faults to accommodate both extensional and shearing phases (Fig. 1). The field exhibits a variety of fault dimensions located beneath a four-way dipping anticline structure. Large active faults extend all the way to the seabed, displaying fault throws of up to 40 meters, which effectively compartmentalizes the reservoir. In addition, smaller faults intersect these larger faults, creating a dense and intricate network of fault systems throughout the area. The faults are crossed by wells both vertically in the overburden and horizontally in the reservoir sections. More precisely, horizontal wells often intersect several large faults that lead to mud losses. Given the complex structural context in the field, we incorporate the fault structure, including the fault core and damage zone, into our geomechanical analysis to evaluate the risk of mud loss during drilling operations.

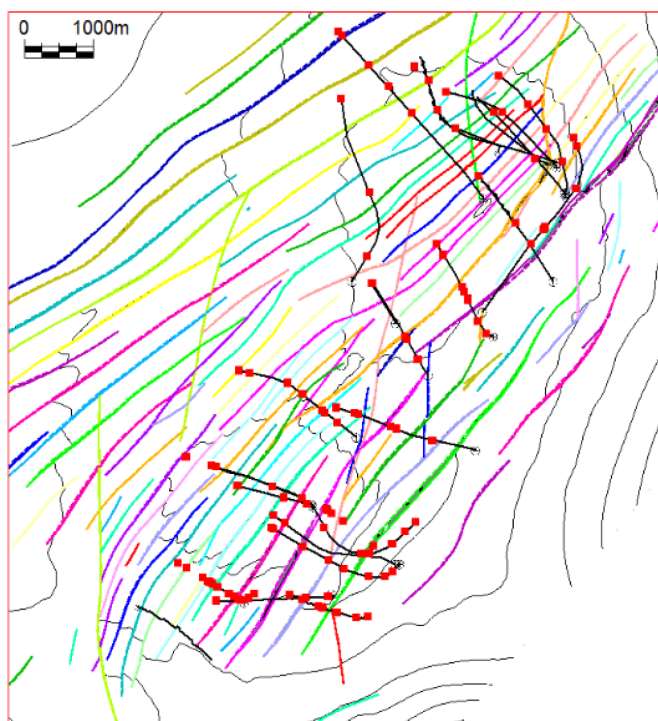


Figure 1—Map view of the heavily faulted area with faults (colorful lines) intersecting well paths (thick black lines) at the top of one reservoir. Mud loss incidents are represented on the well trajectory (red square dots). The bottom right graph is the regional strike slip fault and normal fault accommodation in the center.

Determination of mud losses at proximity of faults

In our study, mud loss incidents have been observed in proximity to interpreted faults, as illustrated in [Fig. 1](#). A comparison of the structural map at the top of the reservoir reveals a spatial correlation with the loss incidents, suggesting that the fault network significantly impacts drilling performance.

To investigate the relationship between faults and mud losses during drilling operations, particularly focusing on fault reactivation, we developed a comprehensive database that systematically extracts mud loss incidents from daily drilling reports. This extensive database documents over 200 mud loss events across more than 25 wells involved in both exploration and development campaigns. The wells represent a diverse range of architectures, including vertical and horizontal trajectories, which provides a robust cross-section of operational conditions. The mud loss incidents occurred at mud weights below the predicted fracture gradients, including the Shmin or FCP, and hypothesized to be linked to fault rather than opening of fractures.

For each recorded loss incident, we meticulously captured the mud weight, ECD, measured depth, and true vertical depth. If mud weight overbalance enters the fault, it may critically influence its stability by reducing the effective stresses on the fault, thereby pushing the Mohr-Coulomb failure envelope closer to the failure criterion than initial conditions. We suspect that mud weight or ECD while drilling may trigger fault reactivation near wellbore, leading to dilatancy and thus increased permeability leading to mud losses, especially in formations with low cohesion or altered stress regimes. Documenting the mud weight during these incidents helps establish a correlation between pressure conditions and the severity of mud losses, which is essential for improving prediction and mitigation strategies. The analyzed incidents took place with static mud weights ranging from 1.10 to 1.20 g/cm³ and dynamic mud weights (ECD) between 1.10 and 1.30 g/cm³. The depths under consideration extend from 2,000 m to 3,500 m TVDRT, covering both shallow overburden intervals and deeper reservoir sections ([Fig. 2](#)).

A second database has been created to characterize the attributes of faults that intersect the same wells previously analyzed for mud loss events ([Fig. 1](#)). The key attributes extracted include dip angle and associated lithology. These attributes were mainly obtained from 3D seismic interpretations and the master logs (lithologs) of the studied wells. Only the large faults, which are sufficiently resolved in the seismic cross sections and well logs, were included in this analysis. Smaller faults and fractures typically fall below the resolution limit of seismic imaging making it difficult to assess their characteristics without downhole imagery, which were not done for the current development wells. The major faults identified extend vertically for several hundred meters up to the seabed, indicating that they are still active today.

The attributes of over 170 faults are compiled in a database, including the intersecting depth with the well, the dip angle and the lithology of the contact. For each fault, the depth along the well trajectory was recorded, including any intersections that extend beyond the total depth (TD). Dip angles were estimated from seismic interpretations. The dip angles range from 53 to 80 degrees, with an average of 69 degrees, which is typical for normal fault systems found in extensional tectonic environments. Additionally, the lithology at and near the fault contact was classified into three categories: shale in contact with shale, shale in contact with sandstone, and sandstone in contact with sandstone. These lithological contrasts are known to affect the mechanical behavior of faults as well as fluid migration.

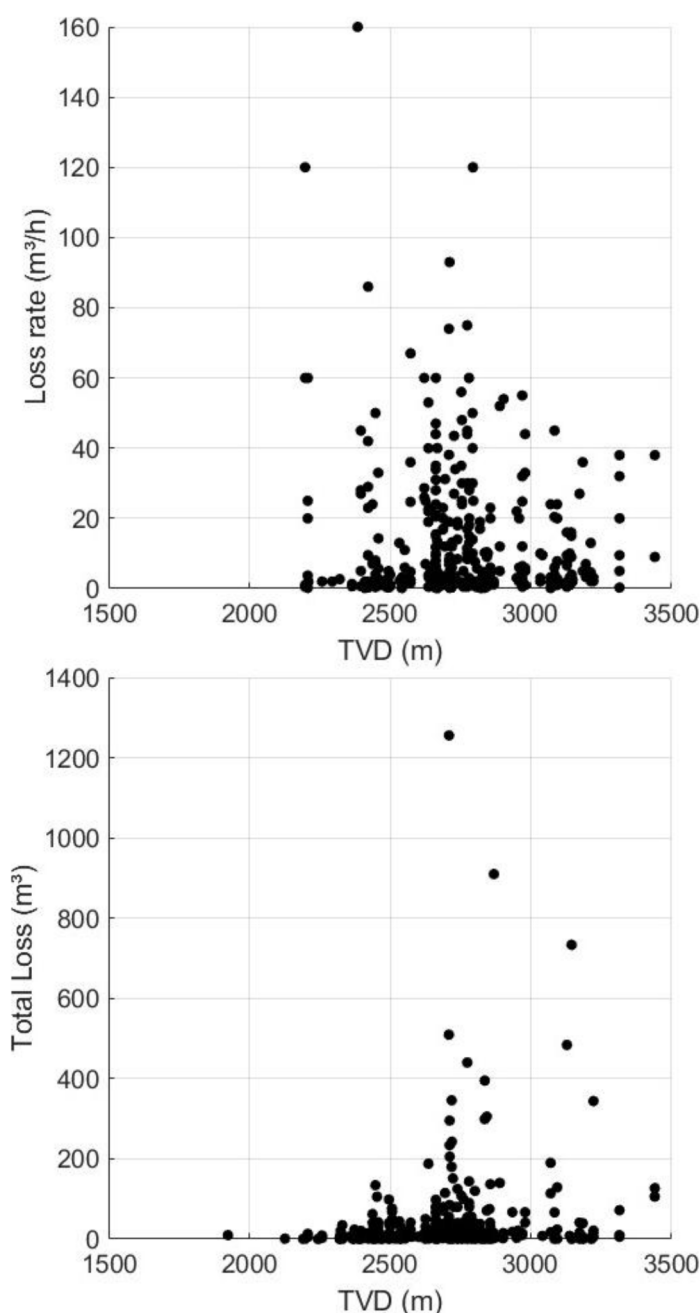


Figure 2—Flow rate (top) and total volume (bottom) of mud losses in the field according to the vertical depth (TVDRT) along the well trajectories

Once the datasets are built, we intersected the mud loss incident dataset and the fault structure dataset based on the spatial correlation of the mud loss measured depth and fault location along well trajectory. To address uncertainties related to wellbore trajectory and seismic fault mapping, the correlation method incorporated a broader view of fault structure, allowing to account for the damage zone that typically surrounds the main fault core. This zone can vary from a few meters to several tens of meters, depending on fault maturity and lithological context, and is often associated with mechanical weakness, making it a likely site for mud losses. Analyses of the dataset indicated that mud loss events frequently occurred within a specific range of fault positions. Based on these observations, a spatial tolerance window of approximately ± 200 meters along the well (mMD) was established. This window accounts for both positional uncertainties of the fault and the extent of structural influence surrounding the faults and was utilized to guide the correlation process (see Fig. 3).



Figure 3—Mud loss events nearby complex faults interpreted within an interval surrounding the fault core: mud loss at the fault core (Case 1); past the fault core (Case 2); ahead of the fault core (Case 3).

A conceptual framework was created to depict three representative scenarios: mud loss occurring directly at the fault plane, beyond the fault, or in advance of it (Fig. 3). In each scenario, the distance between the mud loss point and the fault trace was considered, recognizing both the spatial variability of geological features and the inherent uncertainties in subsurface interpretations. This framework effectively captures the spectrum of possible interactions between drilling-induced fluid losses and fault-related structures.

Geomechanical assessment along well paths

This analysis sought to understand why mud losses often occur near faults during drilling and whether these losses are connected to fault reactivation. We used the Mohr-Coulomb failure criterion to predict fault reactivation by comparing the surrounding stress to the impact of drilling mud. For each well, 1D stress models were computed to illustrate stress conditions and use it in the Mohr-Coulomb analysis.

For each well, we derive the in-situ stress conditions using a simplified approach to build the 1D Mechanical Earth Model (1D MEM), based on the field burial trends. These models include the overburden stress, minimum horizontal stress and the pore pressure (Fig. 4). In the current field, the pore pressure is hydrostatic both in cap rock and in reservoirs at virgin conditions. For each development we compute the overburden stress is calculated from a density trend derived from RHOB logs of the E&A wells. The minimum horizontal stress, referred as S_{hmin} , is derived from the effective stress ratio method calibrated on available ELDT and LOT, (Zoback et al., 2003; Su et al., 2022), and supported by previous 1D models developed in the field and by direct stress measurements from wells:

$$S_{hmin} = k_0(S_v - P_p) + P_p, \quad (1)$$

where S_{hmin} is the minimum horizontal stress, k_0 the effective stress ratio, S_v the vertical stress and P_p the formation pressure.

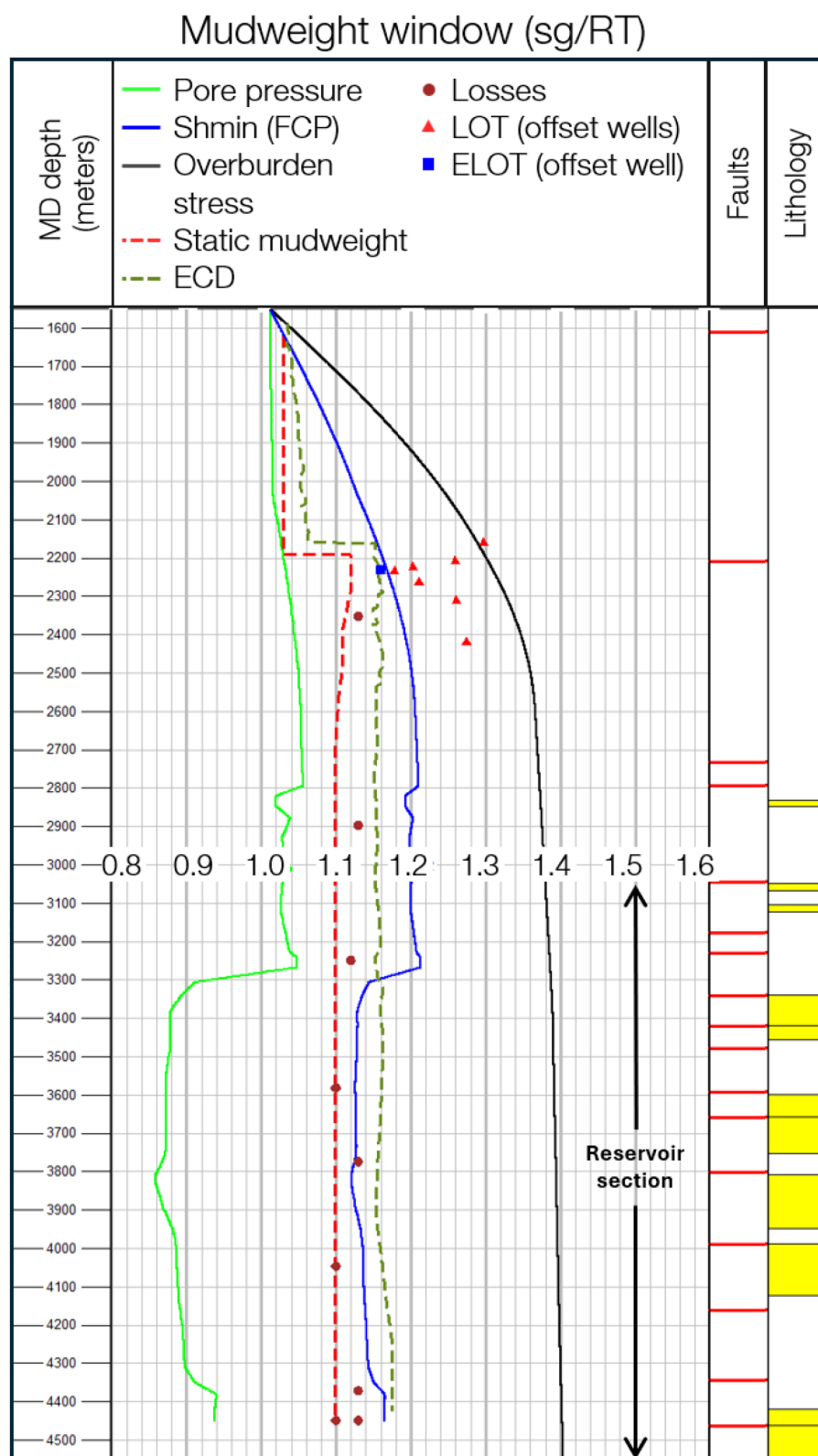


Figure 4—Example of a postmortem 1D geomechanical model (1DMEM) of a slanted well (near vertical in overburden and horizontal in reservoirs) that reflects field conditions and is akin to other 1DMEMs utilized in the field. The depth is measured depth (MD) and principal stresses are illustrated (Pp in green, Shmin in blue, and Sv in black), as well as mud weight used and ECD while drilling (respectively in dashed red and dashed green), loss events indicated by red dots and Leak-Off Test (LOT) results from offset wells represented by red triangles. Faults intersected during drilling are highlighted with red lines in the right panel. The right panel presents a simplified lithological column, indicating sandy intervals in yellow and shaly intervals in white.

To anticipate mud losses caused by fault reactivation during drilling, we employed the Mohr-Coulomb failure criterion. This model evaluates the relationship between shear (τ) and normal (σ_n) stresses acting on a fault, as well as the fault cohesion (C) and friction angle (ϕ), which are critical factors influencing the stability of the fault. In our analysis, the fluid pressure is initially considered to be the in-situ pore pressure; however, we examine the effects of both static and dynamic mud weight on the fault as effective fluid pressure. The effective normal stress is reduced by the fluid pressure, which can significantly weaken the stability of the fault. Specifically, higher mud weights increase the fluid pressure in the formation, thereby reducing the effective normal stress acting on the fault and raising the risk of failure. Consequently, a careful balance of mud weight is critical to prevent fault reactivation and subsequent mud losses during drilling.

$$\sigma'_n = \frac{S_v + S_{hmin}}{2} - P_p \quad (2)$$

Where σ'_n is the effective normal stress, S_{hmin} is the minimum horizontal stress, S_v the vertical stress and P_p the formation pressure.

$$\tau = C + \sigma'_n \tan(\phi) \quad (3)$$

Where τ is the shear stress, σ'_n is the effective normal stress, C is the fault cohesion and ϕ friction angle of the fault. Both the cohesion and friction angle are based on values from intact rock, which are derived from a general compaction trend. Both properties are derived from a computed P-wave slowness trend that corresponds with DTC well log data and reflects a normal compaction trend consistent with the sedimentary deposition observed in the field. This DTC trend is influenced by the burial depth in meters and the average water depth of 1,500 meters:

$$DTC\left(\frac{\mu s}{ft}\right) = 70 + 145 \exp\left(-\frac{burial(m)^{1.05}}{1500}\right) \quad (4)$$

We define the friction angle (ϕ), the cohesion (C) and the UCS of intact rock and estimate these parameters with in-house equations. Initially, no assumptions were made on the fault cohesion and friction angle, instead we determined them based on the intact rock properties and the results.

If reactivation near wellbore occurs while drilling, the effective stresses depend on the static and/or the dynamic mud weight. Therefore, we calculate the effective normal and shear stresses according to the static mud weight and the dynamic equivalent circulating density (ECD).

Results

In Fig. 5, a notable increase in mud loss is observed when the drilling mud weight surpasses the pore pressure of the surrounding formations. The volume of mud loss escalates sharply once the exceeding fluid pressure, defined as the difference between the mud weight and the pore pressure (P_p), exceeds a specific threshold. Mud loss incidents occur more frequently when the exceeding pressure reaches 0.1 sg/RT for dynamic mud weight. Consequently, these findings indicate a clear correlation between the volume of mud weight utilized during drilling and the occurrence of mud loss incidents.

The histogram in Fig. 5 depicts the distribution of mud loss events in relation to fault proximity, revealing a pronounced peak at approximately -52 meters, which signifies a notable clustering of mud loss occurrences near this offset. The median distance between mud loss events and the fault is 18 meters, indicating that half of the recorded events occurred within this interval. This central tendency, in conjunction with the skewed distribution demonstrated by an elevated mean loss volume, suggests a spatial correlation between fault structures and mud loss behavior. Additionally, the map in Fig. 1 further illustrates that these mud loss incidents frequently occur near faults, reinforcing the observed relationship between fault locations and the frequency of mud loss events.

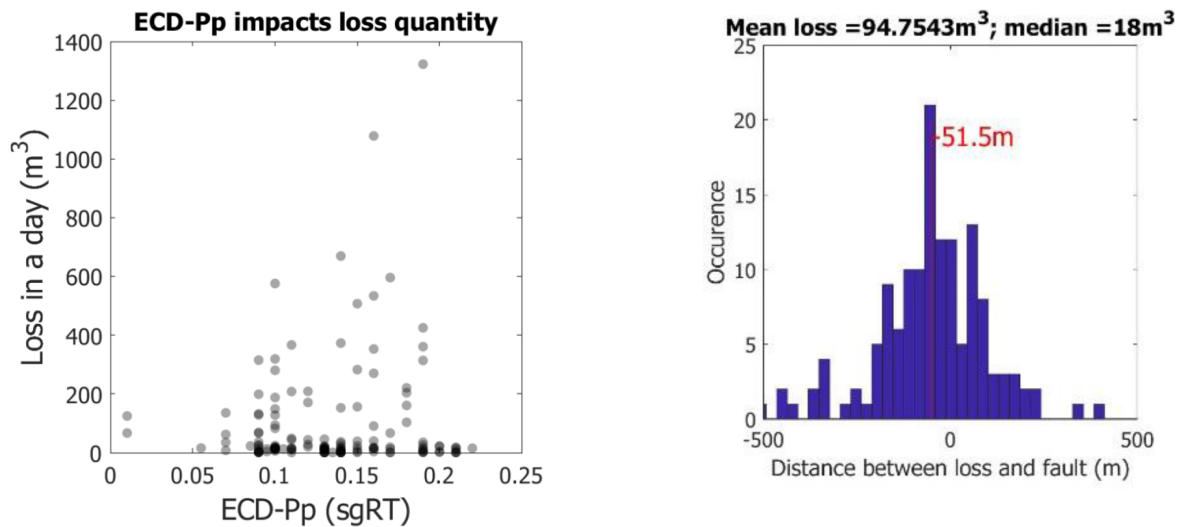


Figure 5—Left: Volume lost in formation according to the exceeding fluid pressure in the fault compared to initial conditions, considering the ECD and the initial pore pressure. Right: Distribution of loss according to the proximity of fault cores.

Our findings, presented on Fig. 6, provide a simplified yet precise visualization of fault reactivation potential under varying stress conditions. Rather than presenting all 170 fault Mohr-Coulomb circles individually, we chose to highlight the critical points on these circles that correspond to the specific dip angles found in our database. This approach simplifies visualization while still providing essential information about the relationship between fault angles and critical conditions. The initial conditions approach the failure criteria when considering a cohesionless fault with a friction angle of 30 degrees. Under these circumstances, the faults are particularly prone to reactivation due to increased fluid pressure during drilling campaigns. As illustrated in Fig. 6, the exceeding fluid pressure driven by mud weight reduces the effective stress on the faults, evident from the shift of the Mohr-Coulomb circles to lower effective normal stresses. Consequently, this indicates that the mud weight used while drilling contributes to fault reactivation, potentially leading to incidents of mud loss.

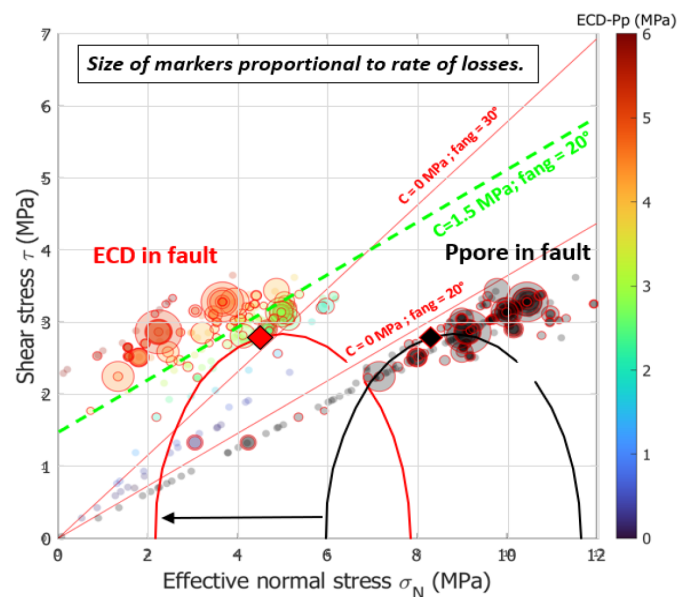


Figure 6—Mohr Diagram with effective normal and shear stresses for faults in our studied field. Losses are analyzed for fluid pressure prior to drilling (P_p in fault, in black markers), and corresponding stress state considering ECD pressurizes the fault (colored dots according to color scale).

We did not make any initial assumptions regarding the friction angle and cohesion, as our aim was to derive fault properties directly from the data. Our findings align with the expected friction angles for siliciclastic rocks, which typically range between 20 and 30 degrees (Fig. 6). Moreover, when considering the equivalent circulating density (ECD), we interpret a cohesion of approximately 1.5 MPa (Fig. 6). This estimated cohesion is consistent with half the cohesion of intact rocks as determined from sonic well logs (Fig. 4).

Discussion

The findings of this study emphasize the importance of evaluating fault reactivation potential when assessing the risk of mud losses during drilling operations. The difference between the equivalent circulating density (ECD) and the initial pore pressure (P_p) emerges as a critical factor in predicting fault instability. Initially, faults in the field studied are close to the failure envelope, particularly when modeled with a friction angle of 20 degrees and no cohesion (Fig. 6). Under these conditions, the faults are within 1 MPa of failure, indicating a high sensitivity to changes in fluid pressure. When ECD is high while drilling, it reduces the effective normal stress acting on the fault, thereby pushing the stress state beyond the failure criterion and triggering fault reactivation. This reactivation is often accompanied by significant mud losses, although not all faults that meet the failure criteria result in fluid loss, suggesting that additional factors such as fault permeability and lithological heterogeneity influence the outcome. Our findings align with previous research demonstrating that elevated equivalent circulating density (ECD) induces fault reactivation and associated losses while drilling (Chan et al., 2019; Brem et al., 2019).

In this study, we propose a new operational criterion with a more precise and optimistic assessment of the friction angles and cohesion of faults in comparison to cohesionless faults. This distinction is critical, as it allows for a better understanding of the mechanical behavior of the faults under various conditions. In the field studied, many of the faults are either currently active or at the onset of failure, highlighting their sensitivity to operational changes (Fig. 6). Notably, the most significant losses were observed when accounting for the equivalent circulating density (ECD) within these faults, particularly when the cohesion was set at 1.5 MPa and the friction angle was 20 degrees. Our friction angle corresponds to an approximate friction coefficient of 0.4, which agrees with previous studies (Chan et al. 2019; Brem et al., 2019). However, our findings demonstrate that faults are cohesive with a cohesion of 1.5 MPa, which is approximately one-third of the cohesion of intact rock (Fig. 6). In contrast, previous studies assumed cohesionless faults in similar geological contexts (Chan et al. 2019; Brem et al., 2019). Our updated criteria are already implemented in pre-drill studies and include recommendations to limit ECD to prevent fault reactivation. By adhering to these established guidelines, operators can effectively reduce the risk of inducing instability while optimizing drilling performance (Abd Rahim et al., 2019).

Failure conditions are defined by effective normal and shear stresses but also lithology at the fault contact. Fig. 7 provides a detailed view of fault behavior by separating Mohr diagrams according to the lithology at the fault contact. The lithological classification reveals that shale/sandstone contacts are particularly prone to fault reactivation near wellbore, with 85% of faults meeting the failure criteria and correlating with mud loss events. Sandstone/sandstone contacts show a moderate risk, with 68% of faults reactivating, while shale/shale contacts, though less frequently reactivated (50%), are associated with the largest volumes of mud loss. These observations suggest that lithological contrasts at fault interfaces amplify mechanical discontinuities, increasing the susceptibility to mud loss. The presence of a shale/sandstone contact is associated with both fault reactivation and potentially more high-volume losses. Chan et al. (2022) previously examined the influence of lithology on mud loss risk, reporting that sandstone formations are associated with a higher incidence of mud loss events. Our results agree with an increase likelihood of mud losses in front of sandstones, more precisely when fault contacts are either shale/sandstone or sandstone/sandstone contacts. While these fault contacts increase the occurrence of mud losses, it is important to note that these incidents

tend to be less voluminous (Fig. 7). We noted that the highest volume of mud loss occurred near faults with a shale/shale contact (Fig. 7). This may result from a different fault structure in overburden shales versus near the sandstone reservoirs and the ability of the mud to provide sealing effect. Fault contacts composed of partially or totally with sandstones, the mud losses might take the form of seepage through the sandy gouge, or directly in the sand formation, that may be cured or tempered by LCM packages or with mudcake effect. Conversely, the voluminous losses in the overburden faults (shale to shale contacts) could be related to leaking pathways towards fracture networks around the fault core, below to seismic resolution, which could not be cured with the mud particles.

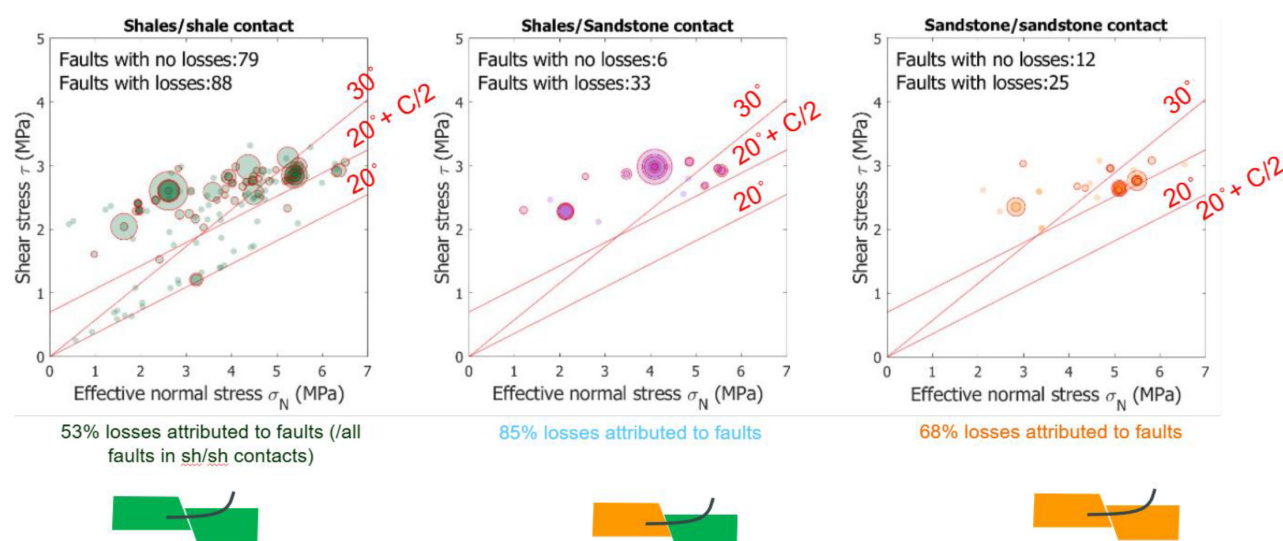


Figure 7—Mohr Diagram with effective normal and shear stresses for faults in our studied field separated by lithology at the fault contact.

Overall, the interplay between fluid pressure, fault geometry, and lithological context is central to understanding and mitigating mud loss risks, as previously highlighted (Brem et al., 2019; Chan et al., 2022). The use of $ECD - P_p$ as a predictive metric, combined with lithology-based fault classification, offers a practical framework for pre-drill risk assessment and operational planning in faulted reservoirs. This approach enables more accurate predictions of mud loss incidents and supports the development of safer and more efficient drilling strategies.

Conclusions

We presented a simple model based on the Mohr-Coulomb failure criteria to predict mud losses in front of normal faults for optimizing drilling performance. Indeed, common approaches often fail to predict mud losses ahead of drilling, such as solely using (Shmin) as an indicator of the onset of mud losses or Mohr-Coulomb approaches with no cohesion. Instead, we adapted the Mohr-Coulomb analysis to more precisely predict mud losses using in-situ stresses derived from 1D Mechanical Earth Models. Firstly, we did not impose specific mechanical properties (cohesion, friction angle) or fault permeability. Secondly, we considered that the equivalent circulating density (ECD) represents the fluid pressure on the fault and decreases the effective normal stress on the fault. In a Mohr diagram, our results allowed us to interpret failure lines defined by half the intact rock cohesion (1.5 MPa) and friction angles between 20 – 30 degrees, consistent with properties in the field. The data indicates that high $ECD - P_p$ leads to fault reactivation and significant mud losses. After analyzing the lithology present at the fault contact, we observed a strong correlation between the risk of mud loss when the criteria are met and lithology. We implemented new criteria that included a small cohesion and a friction angle dependent on the lithology in recent drilling

operations, which succeeded in reducing mud loss incidents. The way forward involves analyzing more recent wells drilled through depleted reservoirs, considering the reservoir stress path is impacted by the proximity of faults and controls Shmin. Furthermore, we also aim to apply our method to several fields to enhance its applicability and effectiveness.

Acknowledgement

The authors would like to thank the affiliate of TotalEnergies and HQ management for their support and permission to publish the results in this paper.

References

- Abd Rahim, M. H., Chan, A. W., Brem, A. G., Seli, P., and M. F. Khodaverdian. 2019. Overcoming Subsurface and Batch Drilling Challenges in a Heavily Faulted Deepwater Environment. Paper presented at the 53rd U.S. Rock Mechanics/Geomechanics Symposium, New York City, New York, June 2019.
- Barton C. A., Zoback M. D., Moos D. 1995. Fluid flow along potentially active faults in crystalline rock. *Geology*, **23** (8): 683–686. doi: [https://doi.org/10.1130/0091-7613\(1995\)023<0683:FFAPAF>2.3.CO;2](https://doi.org/10.1130/0091-7613(1995)023<0683:FFAPAF>2.3.CO;2)
- Brem, A. G., Abd Rahim, M. H., Zhang, T., and A. W. Chan. 2019. How Strong Is Your Fault?. Paper presented at the 53rd U.S. Rock Mechanics/Geomechanics Symposium, New York City, New York, June 2019. ARMA-2019-0297
- Caine, J. S., Evans, P. J. and Forster, C. B. 1996. Fault zone architecture and permeability structure. *Geology*, **24**(11), 1025–1028. doi: [https://doi.org/10.1130/0091-7613\(1996\)024<1025:FZAAPS>2.3.CO;2](https://doi.org/10.1130/0091-7613(1996)024<1025:FZAAPS>2.3.CO;2)
- Chan, A. W., Brem, A. G., and M. H. Abd Rahim. 2019. Lost Circulations Due to Fault Reactivation and the Implications on Stress Characterization. Paper presented at the 53rd U.S. Rock Mechanics/Geomechanics Symposium, 2019. ARMA-2019-0492
- Chan, A. W., Brem, A. G., Abd Rahim, M. H., Numpang, A., & Chong, S. 2022. The four pillars for de-risking fluid loss potential along fault damage zone: a framework for well designs and drilling operations. In ARMA US Rock Mechanics/Geomechanics Symposium (pp. ARMA-2022). ARMA-2022-0197
- Frash, L. P., and James W. C. 2017. Experimental Measurement of Fracture Permeability at Reservoir Conditions in Utica and Marcellus Shale. Paper presented at the SPE/AAPG/SEG Unconventional Resources Technology Conference, Austin, Texas, USA. URTEC-2666764-MS
- Guglielmi Y., Nussbaum C., Cappa F., De Barros L., Rutqvist J. 2021. Field-scale fault reactivation experiments by fluid injection highlight aseismic leakage in caprock analogs: Implications for CO2 sequestration, *International Journal of Greenhouse Gas Control*. **111**, 103471. <https://doi.org/10.1016/j.ijggc.2021.103471>
- Hyman, J. D., Jiménez-Martínez, J., Viswanathan, H. S., Carey, J. W., Porter, M. L., Rougier, E., ... & Makedonska, N. 2016. Understanding hydraulic fracturing: a multi-scale problem. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, **374**(2078), 20150426. <https://doi.org/10.1098/rsta.2015.0426>
- Sibson, R. H. 1986. Earthquakes and Rock Deformation in Crustal Fault Zones" *Annual Review of Earth and Planetary Sciences*, vol. **14**, p. 149, doi:10.1146/annurev.ea.14.050186.001053.
- Sibson, R. H. 1977. Fault rocks and fault mechanisms. *Journal of the Geological Society*, **133** (3): 191–213. doi: <https://doi.org/10.1144/gsjgs.133.3.0191>
- Su K., Onaisi A., Vidal-Gilbert S. 2022. Comparison of different approaches of 1D Mechanical Earth Model used in geomechanics. 56th US Rock Mechanics Geomechanics Symposium. ARMA-2022-0749
- Zoback MD, Barton CA., Brudy M., Castillo DA., Finkbeinerb T., Grollimundb B.R., Moosb D.B., Peskab P., Wardb C.D., Wipruth D.J.. 2003. Determination of stress orientation and magnitude in deep wells, *International Journal of Rock Mechanics & Mining Sciences* **40** 1049–1076. <https://doi.org/10.1016/j.ijrmms.2003.07.001>